

Assessment of Marginal Emissions Factor in Power Systems Under Ramp-Rate Constraints

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Abstract—Reducing carbon emissions in the power sector is critical for transitioning to a sustainable and low-carbon future. Estimating carbon efficiency of demand side response (DSR) in the power system is an important step towards realizing potential environmental benefits. Marginal emission factor (MEF) is an effective tool for estimating incremental changes in carbon emissions as a result of a change in demand. However, estimation methods currently used for evaluating MEF can be improved upon, specifically, by factoring in the ramp-rate constraint of generators in the fuel cost based merit order dispatch. In this paper is described a new method for MEF assessment under ramp-rate constraints; the method is then compared with two conventional estimations in a British power system. Three fuel price scenarios are used to conduct a sensitivity analysis of MEFs to fuel prices. Conclusions are drawn that can pave the way for future improvements in estimating MEF in power systems.

Index Terms—Carbon emission, fuel prices scenarios, marginal emissions factor, ramp-rate constraint, sensitivity analysis.

I. INTRODUCTION

THE UK Department of Energy and Climate Change's (DECC) carbon plan, released in December 2011, provides a roadmap for transitioning to a low-carbon economy [1]. Under this plan, the power sector—responsible for 27% of carbon emissions and constituting the single largest source of carbon in 2010—must cut emissions to almost zero by 2050.

Proposed strategies for decarbonizing the power sector consider the potential impacts of changes to both the supply-side energy mix and demand-side usage changes. On the supply side, generation will be decarbonized through deployment of renewable energy and nuclear power, and implementation of carbon capture and storage technologies [2]–[5]. On the demand side, electrification of heat and transport will be deployed to reduce carbon emissions in other sectors, and flexible demand will play an important role in future power systems portfolio [6].

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Carbon emissions in the power sector depend on both demand-side energy use and the carbon intensity of generation units. The potential of demand side response (DSR) has been recognized as increasingly important to ensure the future security of electricity supply and to reduce cost and carbon emissions [6]–[8]. One important consideration in appropriate demand management is a proper assessment of the impact of demand-side intervention in carbon reduction, which means that demand-side can play an active role in meeting carbon reduction targets [9]. In order to grasp the impact of DSR, a range of carbon emissions factors have been proposed and published that attempt to evaluate the environmental benefits of DSR [10]–[18].

Carbon emissions factors typically calculate the amount of greenhouse gas emissions caused by energy use [13] based on which we are able to assess how much carbon would be emitted or saved by changing electricity usage in terms of kg carbon dioxide per kWh of power usage (kgCO_2/kWh). Two types of carbon emission factors are commonly used for this assessment: average emission factor (AEF) and marginal emission factor (MEF). AEF is defined as the ratio of CO_2 emitted to electricity generated, which represents the average kgCO_2/kWh of electricity consumed at the point of final consumption, including both carbon emissions from power generators and emissions associated with transmission and distribution losses [19]–[21]. MEF is widely defined as the incremental change in carbon emissions as a result of a change in demand [10]–[18]. This is a useful tool for analyzing fuel cost benefits and emission benefits from DSR.

Current MEF analysis methods do not consider technical properties such as ramp-rate constraints. In practical terms, to expect a fast ramp rate of thermal units, such as in coal-fired power plants, is clearly not possible, which is why the merit order MEF, which takes into account ramp-rate constraints in the dispatch order, is necessary. In past estimations of fuel cost based MEF analysis, the prevailing fuel price was used to form the merit order. However, the fuel cost based order is subject to fluctuations in fossil fuel prices, which, over the past 10 years, have changed significantly. Gas prices, for example, have risen by 125% and the price of coal has increased by 48% in the past decade [22]. Such changes are expected to continue into the future, which is why a proper sensitivity analysis of MEFs is necessary to demonstrate how different fuel prices can necessarily impact these changes in MEF analysis.

In this paper, the conventional fuel cost based merit order method is improved upon by factoring in the impact of ramp-

rate constraints in the MEF estimation. The proposed method in conjunction with two conventional merit order methods (the fuel cost based MEF and the utilization based MEF) are applied in two scenarios of the British power system (typical winter demand and typical summer demand) to illustrate the potential differences between AEF and MEF in Great Britain (GB). Three fuel price scenarios are also used to examine how the MEF changes with fuel prices.

II. LITERATURE REVIEW AND PROBLEM STATEMENT

A. Average Emissions Factor (AEF)

AEF is efficient for the purposes of evaluating emission footprints in the power sector; it is also capable of monitoring and reporting overall carbon emissions at a system level and on an average basis. However, the problem with the AEF is that it does not reflect the marginal effect of a change in demand. AEF is not appropriate for assessing marginal carbon benefits from the perspective of distribution side response (DS).

MEF, on the other hand, is considered an appropriate instrument for evaluating the marginal emission change as a result of a change in demand [9]–[18]. Studies on quantification of such marginal change can be classified into three categories: single marginal plant approach, merit order approach, and empirical approach.

B. Single Marginal Plant MEF

The intent of MEF is to reflect the actual change in emissions that would result from a small, but sustained change in electricity consumption. It is mainly determined by the marginal plants whose output increases or decreases based on a sustained, but marginal change in the system's demand. The Valuation of Energy Use and Greenhouse Gas Emissions for Appraisal issued by the UK Department of Energy & Climate Change (DECC) estimates the marginal plant to be a single combined cycle gas turbine (CCGT) plant. This is based on the assumption that the CCGT plant is and will reflect a significant part of the marginal plants. The MEF is then estimated to be the effect of a CCGT plant plus a fixed rate of system loss (see Table I).

TABLE I
MEFs BY U. K. GOVERNMENT [22]

Year	Generation-Side (kgCO ₂ e/kWh)	Assumed Grid Losses	Consumption-Side (kgCO ₂ e/kWh)
2010	0.3464	7.50%	0.3723
2011	0.3400	7.50%	0.3655
2012	0.3333	7.50%	0.3583
2013	0.3261	7.50%	0.3505
2014	0.3184	7.50%	0.3422

CCGT plant is an important marginal plant and has a significant impact on MEF. However, moving forward, other plants such as the coal-fired and low carbon technologies are also accountable for marginal impacts. There are reasons to believe that MEF may not be determined by a CCGT unit only. For example, there are policies that encourage carbon-intensive generators to work on the margin, and many efforts are being made to make the coal-fired more flexible.

C. Empirical Approach MEF

The empirical approach uses linear regression to estimate the average response of each generation technology class; it disaggregates historical data of generator dispatch and carbon emissions in order to examine the influence of marginal effects. It has the advantage of being able to incorporate technical properties of the power system, e.g., grid constraints and power losses. In [11], Hawkes took into account structural changes in the mix of generators, and estimated the UK's long-run MEF based on minimal fuel-cost criteria. It was concluded that MEF would be approximately 0.26–0.53 kg CO₂/kWh for the coming decade in the British power system [11]. Nevertheless, the accuracy of this methodology depends primarily on the historical dispatch data, and an estimation error could have occurred due to inadequate data.

D. Merit Order Based MEF

In power generation, power plants are classified into two types: base-load power plant and marginal power plant. A base-load power plant is a power station that provides a continuous electricity supply to customers, with the characteristic of slow demand respond. It is not capable of meeting flexible demand or even respond to DSR. A marginal power plant, on the other hand, can adjust its output quickly to balance the changing demand, and is capable and suitable for taking the DSR.

In order to reflect marginal impacts of the DSR on carbon emissions, studies on MEF estimations assume a merit order of the marginal plants that is to be triggered according to some specific criteria, e.g., fuel cost and utilization level of generators.

For a utilization-based merit order, Bettle and Pout used the historical data from the power systems of England and Wales in 2000 to develop a utilization-based merit order, with the highest utilization as the lowest in the merit order [18]. The operational data of the power plants was used to decide generator loading levels on a half-hourly basis. Generators were then ranked according to their loading percentage and a utilization-based merit order was then formed for MEF estimation [18]. It was furthermore demonstrated that when electricity demand peaked in winter, the general order was nuclear, followed by CCGT, oil, and coal. Examining the differences between the AEF and the MEF in the case of England and Wales, MEF was seen to be up to 50% higher than the AEF, and using AEF in marginal estimations was likely to seriously underestimate short-term carbon savings.

For a cost-based order, Voorspools and D'Haeseleer activated generators in the power system according to their fuel prices, in which generators with the highest running cost were triggered first for power balancing [16], [17]. Based on 1997 data from the Belgian power system, these studies showed that the generators in Belgium were placed in fuel-cost merit order with the most expensive generator (open cycle gas turbine, OCGT) as the first to respond to DSR. In both these studies, MEF was estimated to be around 1000 g CO₂ per additional kWh, after applying some additional demand patterns to the existing overall electricity demand. Marnay

and Fisher used another cost-based order to calculate MEFs for the Californian electricity network [12]. Here, a model called Elfin or the Electric Utility Financial & Production Cost Model was employed to consider generator response to demand changes. The model is based on electricity production cost, and researchers in this work concluded that MEFs could range from 25% to over 200% higher than the corresponding AEF [12].

Let us assume a merit order of response is a plausible way of assessing MEF in a power system. However, it is important to consider also whether a generator is suitable for balancing the DSR, which can be only determined not by its fuel cost or utilization level, but also by its own specific constraints. A generator has many constraints. For example, it has upper and lower limits, typically referred to as the limits of sustained generation within a certain range of values. The generator also has emissions constraints, i.e., the total limit of emissions ($\text{SO}_x/\text{NO}_x/\text{CO}_2$) under normal operating conditions. Finally, it also has ramp-rate constraints, which denotes the speed at which a generator can increase (ramp up) or decrease (ramp down) generation [24]–[26]. Overall, ramp-rate constraint is considered the most influential in affecting MEF estimation. This is because MEF is an estimation of a generator's marginal response to a change in demand, and the upper bound of the response is decided mainly by the ramp-rate constraint.

Some generators have a very quick ramp rate. For example, a gas-fired generator might be able to ramp up to its maximum output in just 10–15 minutes, compared with hours for a typical coal-fired power plant to reach its full output [24]. Hydro power has high availability and a quick ramp rate, and is considered a balancing source in a power system. The coal-fired power plant has flexibility of changing output to meet changes in demand; however, its flexibility is limited, and the limitation is by and large the ramp rate. In other words, a coal-fired power plant might be triggered according to the assumed criteria to meet the DSR, but it is possible that it cannot adjust its output as expected due to the ramp-rate constraint. A fast ramping up and down is not plausible for coal-fired generators. Therefore, generators with limited flexibility have to be constrained in MEF estimation, and the effect of ramping constraint should be an important consideration.

III. CONSIDERATION OF FUEL PRICES

A. MEF Subject to Fuel Prices

The fuel cost based approach, which is based on assumption of fuel prices, identifies expensive generators as the marginal plants for MEF estimation. However, existing estimations are generally based on current or historical fuel prices, and therefore lack proper sensitivity analysis for MEF estimation. For example, the fuel costs used in [16], [17] are based on fuel prices in the year 2000, which is out-of-date since significant changes have taken place in both domestic and international fuel markets during the past decade. Therefore, a proper sensitivity analysis is necessary to show how MEF estimations can be affected as a result of changing fuel prices.

B. Past and Future Fuel Prices

The price of fossil fuel has steadily increased in the past two decades. The following three charts show how the real prices of coal, gas, and oil have changed since 1987 [22]. The data presented here are given in terms of index numbers, which is a way of presenting proportionate changes in a common format, and a way to observe changes over time [27]. The base year used here is year 1987, and other data are interpreted as percentage differences to arrive at the index numbers.

As shown in Fig. 1, gas prices fell gradually between 1987 and 2000, and in late 2000, prices were around one-third below the 1987 level. The main reasons for the fall in price up until 2000 were price controls set by regulators and, latterly, the impact of competition [22]. However, after 2000 the price began to rise consistently. Price growth between 2000 and 2005 was relatively smooth, but rose rapidly after 2005 and peaked in 2008.

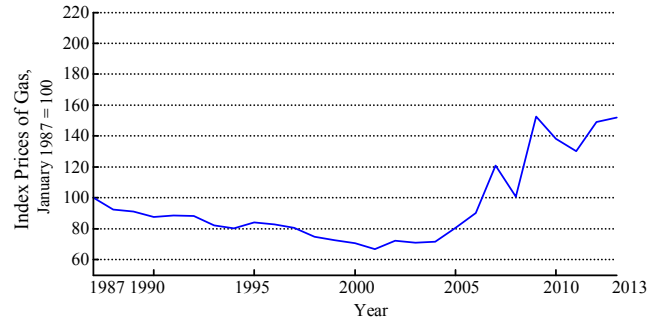


Fig. 1. Index price of gas [22].

As shown in Fig. 2, coal prices display a similar trend as gas: they were down between from 1987 and 2000, and rose afterwards. Coal prices were more stable than gas, having smaller fluctuations. There were some increases in coal prices after 2005, but the evidence does not reflect a big price increase in the near future.

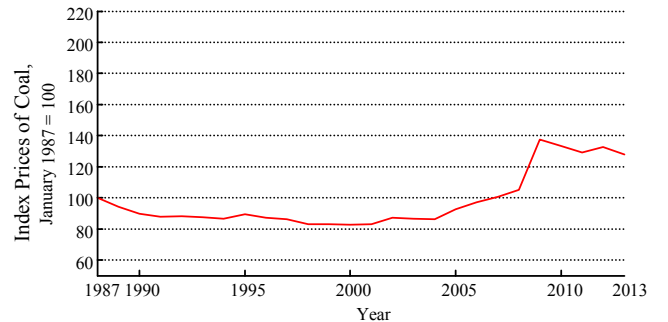


Fig. 2. Index price of coal [22].

The forecasting of fossil fuel prices presents a challenge since it is difficult to manage the uncertainties of determining the price. Moreover, the volatility that comes with economic growth and development of new technologies also contributes to the uncertainties. As such, this paper refrains from forecasting the fissile fuel prices; instead, it will use the reliable projections of fuel prices from the DECC [28]. The key point

of this study is to see how MEFs might change over time with future price scenarios. DECC's projections of coal price and gas price are given in Fig. 3 and Fig. 4.

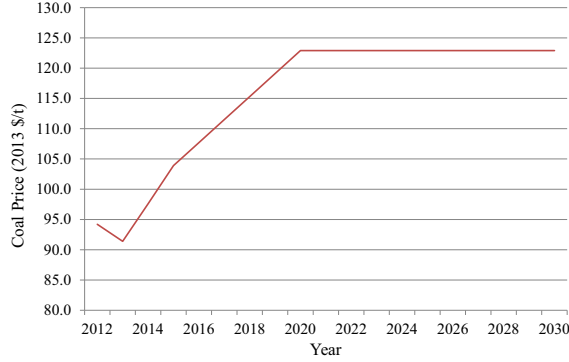


Fig. 3. DECC's coal price projection [28].

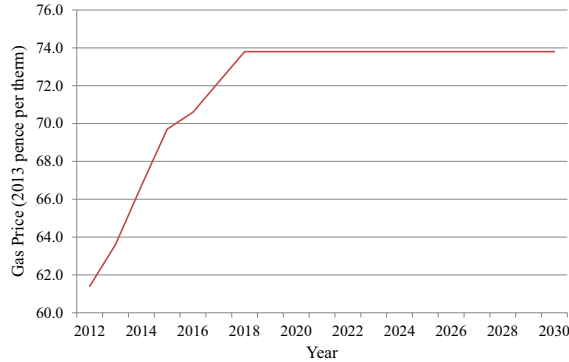


Fig. 4. DECC's gas price projection [28].

IV. MERIT ORDER MODEL WITH RAMP-RATE CONSTRAINT

To grasp the impact of demand side on carbon emissions, several start-up criteria are needed to determine which plants have the priority to meet the demand change in a power system. Based on the assumption that generators with highest fuel costs are desirable for balancing DSR, the generating plants are ranked for dispatching according to the following equation:

$$Rank = \min\{FC_1, FC_2, \dots, FC_i, \dots, FC_N\} \quad (1)$$

$$FC_i = \alpha + \beta \times G_i + \gamma G_i^2 \quad (2)$$

where FC_i refers to fuel cost generator i . α , β , and γ are constant/linear/quadratic coefficient of a quadratic cost curve. G_i refers to the output of generator i . Equation (2) is a quadratic cost function of the generation level G_i . The general principle is to model each generator's cost function independently, but complexity can be reduced by modeling each generation technology as a group, and then by assuming average annual values to each group.

There are, however, various constraints faced by a generator. For a given generator i , it can produce electricity within a certain range of values, which is the upper and lower limit.

$$G_l \leq G_i \leq G_u \quad (3)$$

where G_u refers to the maximum available capacity of generator i . G_l refers to the minimum stable generation of generator i .

Apart from the upper and lower limits, generator i has limits on how quickly it can ramp up and down in order to satisfy the power balance. The modified output power level of the given plant from time t to time $t + 1$ needs to be within ramp-up and ramp-down constraints.

$$G_i^t - G_i^{t+1} \leq RR_i^u, \quad \forall t \quad (4)$$

$$G_i^t - G_i^{t+1} \leq RR_i^d, \quad \forall t \quad (5)$$

$$G_l - G_i^t \leq G_u, \quad \forall t \quad (6)$$

$$G_l \leq G_i^{t-1} \leq G_u, \quad \forall t \quad (7)$$

where G_i^t refers to the output of generator i at period t . G_i^{t-1} refers to the output of generator i at period $t - 1$. G_i^{t+1} refers to the output of generator i at period $t + 1$. RR_i^u refers to ramp-up limit, and RR_i^d refers to ramp-down limit.

V. SCENARIOS AND ASSUMPTIONS IN DEMONSTRATION

A. Scenarios Used for Demonstration

To set the comparisons of MEFs between the proposed method and other methods in an appropriate context, typical winter and summer demands in GB's power systems are used in this study. Briefly, the British power system largely consists of three major groups of power plants: nuclear power plants (13%), CCGT plants (35%), and coal-fired units (35%). As a point of interest, Fig. 5 and Fig. 6 are given to show how demand was actually met by the generators' response during the typical winter demand (Wednesday, 17/11/10) and typical summer demand (Thursday, 10/06/10), respectively [29].

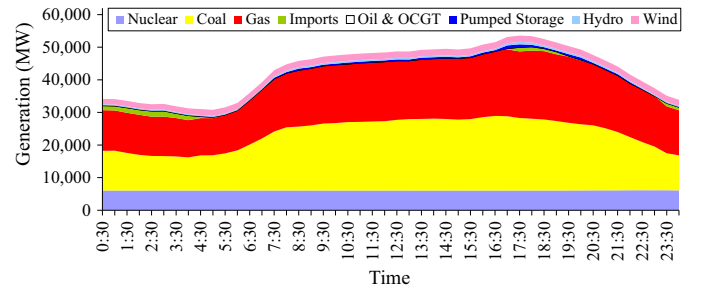


Fig. 5. Typical winter demand: Wednesday, 17/11/2010 [29].

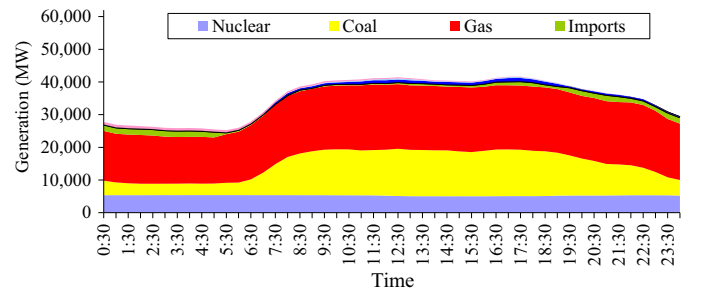


Fig. 6. Typical summer demand: Thursday, 10/06/2010 [29].

B. Assumption of Demand Reduction

Electric power planners expect increased participation in DSR to decrease peak electric power demand by up to 5% over the next 10 years [30]. The relationship between electricity demand reduction and carbon emission is considered to be central to the low-carbon emission policy. It is clear that carbon savings per kWh are not the same with various demand reductions, as different reductions can trigger different selections of generators to respond. As such, the sensitivity of MEF to different levels of demand reduction needs to be examined. In this study, three demand reductions are used to show the effects of different demand reductions on the MEFs: tiny reduction (1%), moderate reduction (5%), and mass reduction (10%). To figure out the MEF changes with time, all hours of the day are assumed to experience reductions at these percentage levels.

C. Assumption of Ramp Rate

For every single generator, ramp rate is one of the basic properties in the generator's design menu. The general principle is to model each generator's ramp rate independently, but complexity can be reduced by modeling each generation technology as a group, and then by assuming a ramp rate for each group. In this study, the ramp rate is carefully selected by analyzing the dispatch profile of each technology and assuming the biggest change of the day to be ramp rate for this technology. For example, in the typical winter scenario, the maximum change of the coal-fired technology was 2145.92 MW that happened in half an hour. So the upper constraint of the ramp rate for the coal-fired plant in the typical winter scenario is chosen as 4292 MW/h. Furthermore, two additional ramp rate scenarios (-5% and $+5\%$) are also considered since the power industry has begun to improve the flexibility of coal-fired technology; it is possible that future coal-fired plants will run more flexibly [31], [32].

VI. CASE STUDY I: TYPICAL WINTER SCENARIO

In order to have a better understanding of marginal carbon emission factors, two typical merit-order approaches (fuel cost based and utilization based) are selected to weigh their merits and drawbacks in MEF assessment. The new MEF approach that takes into account the ramp-rate constraint is also applied in the scenario in order to make a comparison. Fossil fuel prices in this section are based on the data from [13] and sensitivity analysis of MEFs to fuel prices is done in the section that follows.

A. Impacts of Ramp Rate

First, MEFs without technical constraints are estimated in Fig. 7. The results in Fig. 7(a) show that if 1% uniform reduction is applied, the utilization based MEFs are very close to the fuel cost based MEFs in off-peak hours of typical winter demand. The MEFs of off-peak hours are around 50% higher than the system average factors - AEF.

In peak hours, the utilization based MEF is slightly smaller than the cost based MEF, but both are bigger than the AEF.

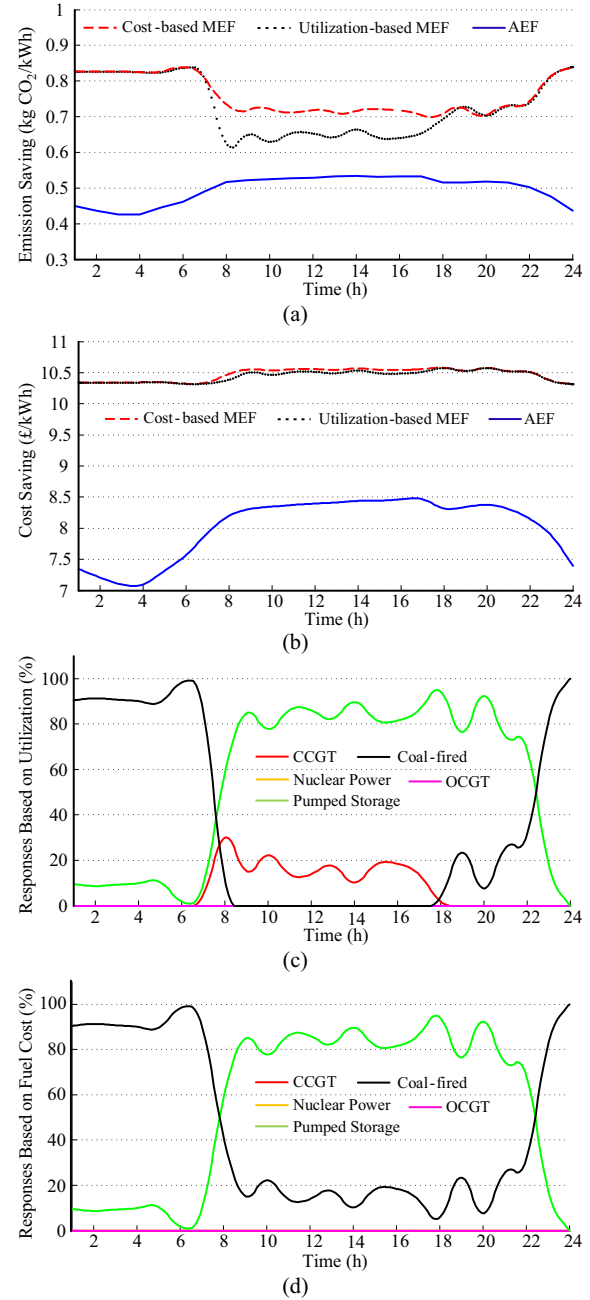


Fig. 7. Typical winter demand, 1% demand reduction.

The result in Fig. 7(b) is a comparison of fuel cost savings among these three factors. The cost savings shown indicate the fuel costs that can be saved when the system demand is reduced. It can be seen that fuel cost savings of the utilization based assumption and cost based assumption are very similar when system demand reduction is considered to be 1%.

The reason for such similarity in cost estimation and difference in MEF estimation is mainly because of the marginal generators assumed. Both the utilization based order and cost based order assume around 80% of demand reductions taken by pump units during the peak hours. However, the utilization based order assumes CCGT units take the remaining 20% of demand reduction, and the cost based order assumes the coal-fired plants are responsible for the remaining 20% (shown in

Fig. 7(c) and Fig. 7(d). Different assumptions lead to small differences in cost savings, as fuel costs for CCGT and the coal-fired plants are almost the same according to the provided data, 10.3174 £/MWh for the coal-fired and 10.0192 £/MWh for the CCGT. But cost based MEFs are that much bigger, as carbon efficiency for the coal-fired (0.85 kg CO₂/kWh) is much higher than the CCGT (0.40 kg CO₂/kWh).

The MEF estimation with consideration of ramp-rate constraints is shown in Fig. 8. Due to the demand reduction considered being insignificant, restrained generators have enough ramping capability to balance the change without changing the merit order. The ramp-rate constraint does not affect the MEF estimation.

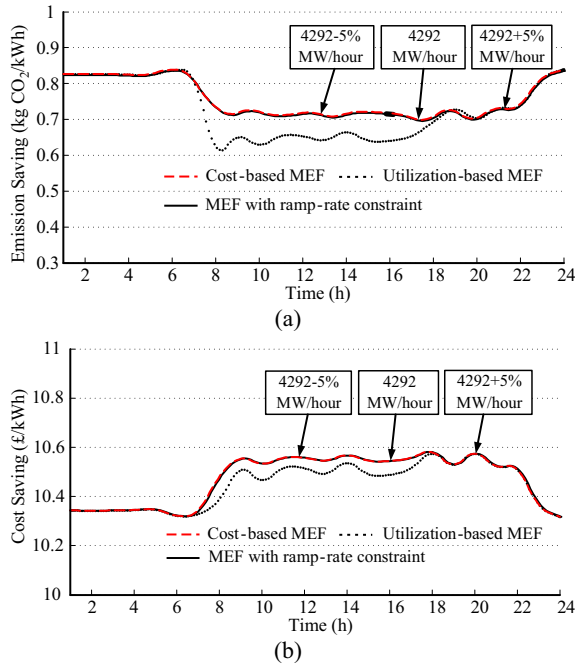


Fig. 8. Typical winter demand, 1% reduction with ramp-rate constraint.

When the demand reduction is increased to 5% (as shown in Fig. 9), the cost based MEF is almost unchanged, but the utilization based MEFs dip to 0.45 kg CO₂/kWh during the peak hours (as shown in Fig. 9(a)). This is because the utilization based MEFs assume the CCGT units to be the marginal plants during the peak hours, which accounts for 80% of the 5% reduction. The cost based MEFs assume the majority of demand change to be taken by the coal-fired plants (Fig. 9(d)). Therefore, during the peak time, the cost based MEF is close to the coal-fired value (0.85 kg CO₂/kWh), and the utilization based MEF is close to the CCGT value (0.4 kg CO₂/kWh). In terms of cost savings, the cost based estimation is slightly higher than the utilization based estimation during the peak hours (as shown in Fig. 9(b)).

However, if considerations of ramp-rate constraint are included in MEF estimation, the results are different. As can be seen in Fig. 10, the MEFs with constraints are lower than the original cost based MEF when the ramp rates are set up to be 4292 MW/h and 4077 MW/h (4292-5%). The reason is that the coal-fired generators are being restrained.

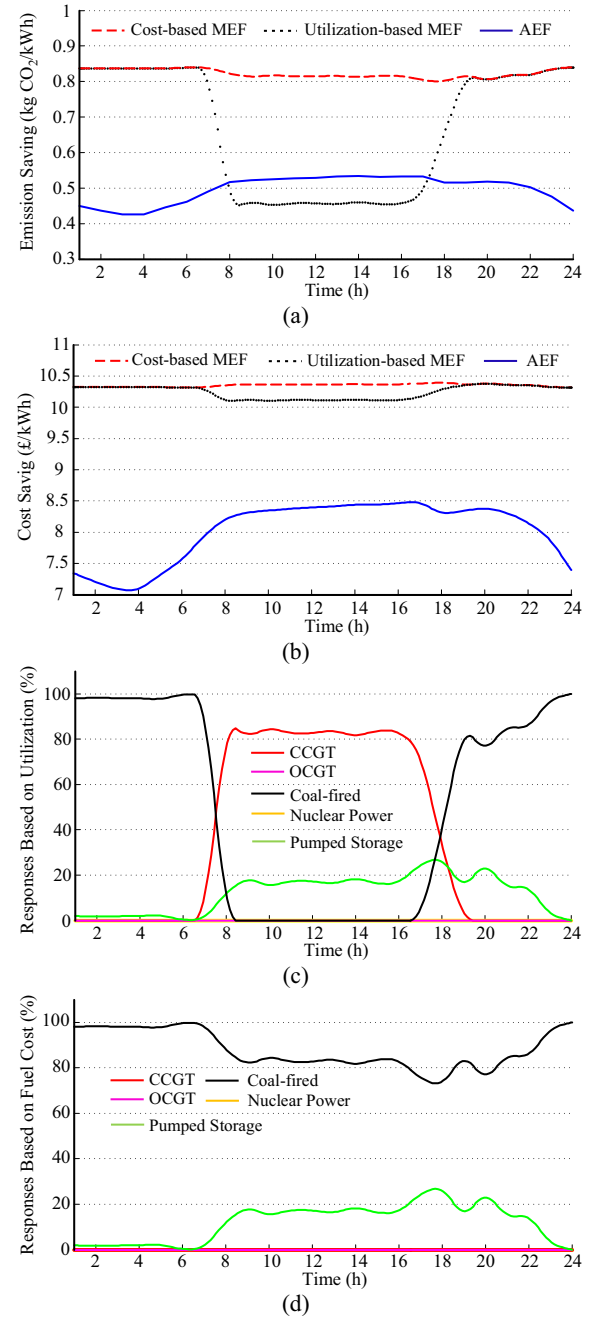


Fig. 9. Typical winter demand, 5% demand reduction.

The coal-fired power plant has limited flexibility of changing output to meet the demand changes. It is not possible to expect a fast ramp-down of coal power generators, even under a cost based merit dispatch. Therefore, when the demand reduction is bigger than the coal-fired plant's ramping capability, the coal-fired plant is not able to take the whole reduction as expected. Part of the demand change has to be transferred to the other generator that is next in the merit order. In this case, the demand changes that are supposed to be totally taken by the coal-fired according to a cost based merit dispatch are shared between the coal-fired and the CCGT units. The CCGT is a much cleaner technology than the coal-fired and the MEFs are thus lowered.

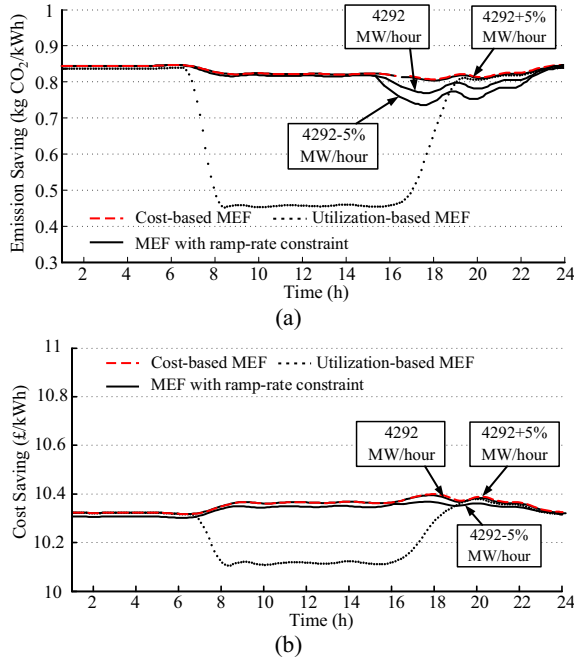


Fig. 10. Typical winter demand, 5% reduction with ramp-rate constraint.

If the demand reduction is increased up to 10% (Fig. 11), both the cost based MEFs and the utilization based MEFs are almost unchanged compared with 5% reduction. However, if the ramp rates are considered (Fig. 12), results of MEFs turn out to be very different. The MEFs with constraints are much lower than the fuel cost based MEFs, even under a cost based merit dispatch. This is due to more demand reduction being transferred to the CCGT units, as 10% is a mass change, and the output of a coal-fired plant is limited by its ramp rate. The coal-fired unit cannot experience a fast enough ramp down in this scenario, thus triggering the CCGT to take part of the reduction. Values of MEFs lie in the range between the coal-fired emissions rate and the CCGT's emissions rate.

From these three levels of demand reductions (1%, 5%, and 10%), it can be seen that the differences between the cost based MEF and the utilization based MEF during the peak hours are mainly due to different marginal generators being assumed. The cost based MEF considers the pump units and the coal-fired plants as responsible for demand reductions, while the utilization based MEF considers pump units and the CCGT. The cost savings of these two MEFs are similar, but emissions savings indicate big differences.

When the demand reduction is small (1%), the generators' ramp rate has little impact on the results of marginal carbon savings as the flexible generator has enough capability to balance the reduction. But if the demand reduction is significant, the ramp rate can have an impact on the MFE estimation, and consideration of ramp-rate constraint becomes indispensable.

B. Different Fuel Prices

In order to see how MEFs might be affected by changing fuel prices, two more price scenarios are used in this section: current fuel prices scenario and future fuel prices scenario.

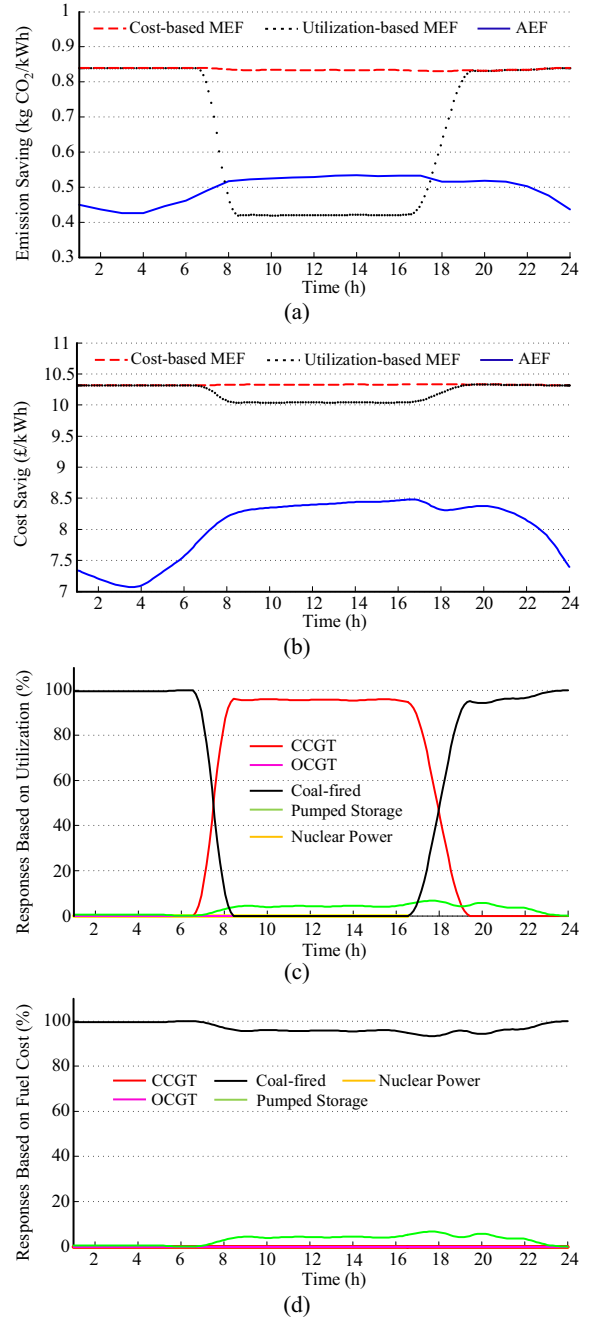


Fig. 11. Typical winter demand, 10% demand reduction.

The current fuel prices scenario is based on the data from DECC [33], which are more up-to-date than previous scenarios. Prices in current scenario can be summarized as: 73 £/MWh for the OCGT, 30 £/MWh for the coal-fired, 48 £/MWh for the CCGT, and 5 £/MWh for the nuclear. The future fuel price scenarios consider the possible increases in fossil fuel prices, and then fuel prices are updated according to the projection. Prices in future scenarios can be summarized as: 91 £/MWh for the OCGT, 39 £/MWh for the coal-fired, 60 £/MWh for the CCGT, and 5 £/MWh for the nuclear plant. Prices in the two scenarios are then used to build the updated fuel cost based merit order, and updated MEFs are thus estimated.

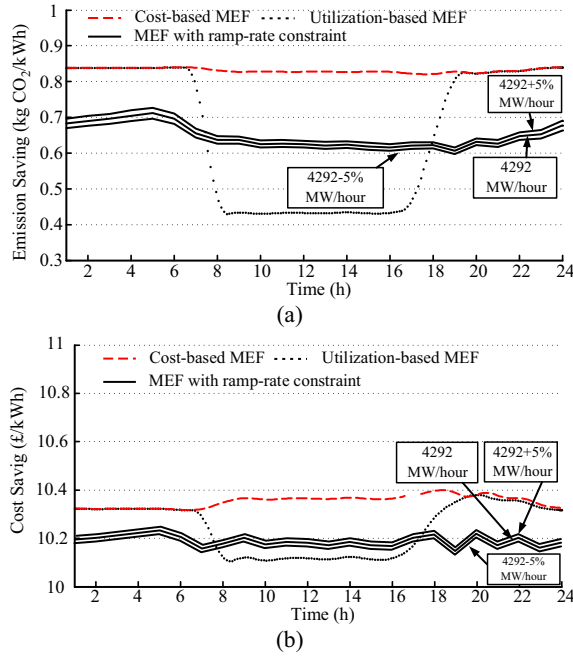


Fig. 12. Typical winter demand, 10% reduction with ramp-rate constraint.

Results in Fig. 13 show that the increase in gas price has fundamentally changed the dispatch order for MEF estimation. The CCGT units are expected to become more active in responding to demand than the coal-fired plants, as the price of gas is more expensive than coal. The coal-fired power plants are not expected to take any demand change at all, as coal is cheaper than gas. The MEFs based on current fuel prices are much smaller than the MEFs based on past fuel prices, because CCGT emits much less carbon emissions than coal-fired units.

Fig. 14 shows very similar results to Fig. 13. It seems that 10% demand change results in similar MEFs to 1% demand change. So a conclusion can be drawn in this case that if the MEF estimation is based on fuel-cost merit order, changes in demand reduction have smaller impact on the results.

When future fuel prices are used to set up the merit order, nothing changes except the potential cost savings. As shown in Fig. 15 and Fig. 16, estimated MEFs are unchanged and generators are responding in the same way as in the current price scenario. But the potential cost savings are higher with future fuel prices because of the increase in price.

Based on these analyses, it can be seen that fuel cost based MEFs are significantly different in the past price scenario and in the current price scenario. This is because gas prices become more expensive than coal and the merit order is fundamentally changed. However, going from the current scenario to future scenario, the MEFs hardly change due to the factor that gas is and will be more expensive than coal. There is no chance for coal to be more expensive than gas in terms of fuel prices. The fuel cost based MEFs are subject to fuel prices, but the order of dispatch will remain stable for a long period of time, which to some extent indicates the necessity of introducing carbon mechanisms to make low-carbon technology more competitive in electricity generation.

On the other hand, in both the current scenario and future

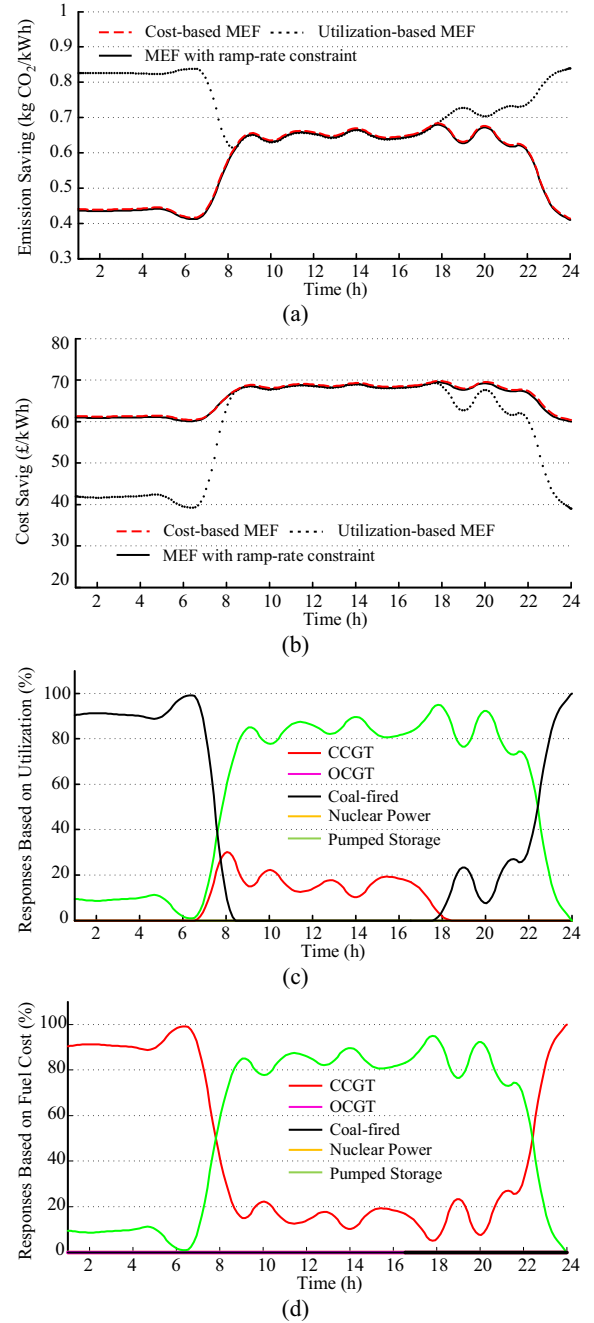


Fig. 13. Typical winter demand, 1% reduction, current fuel prices.

scenario, the coal-fired power plants are not expected to take any DSR. It is true that the coal-fired unit is neither as flexible as the CCGT unit nor as clean as nuclear power. It is a generation technology with limited flexibility. However limited though, its flexibility should be considered in MEF estimation. Actually, the power industry has begun to take another look at how coal can be run more flexibly in the future [31], [32]. Based on this analysis, it can be seen that if the coal-fired could be more flexible, the carbon savings by demand reduction would be much higher than expectations. If the limited flexibility of the coal-fired power plant could be made use of, DSR would be much more effective in reducing carbon emissions.

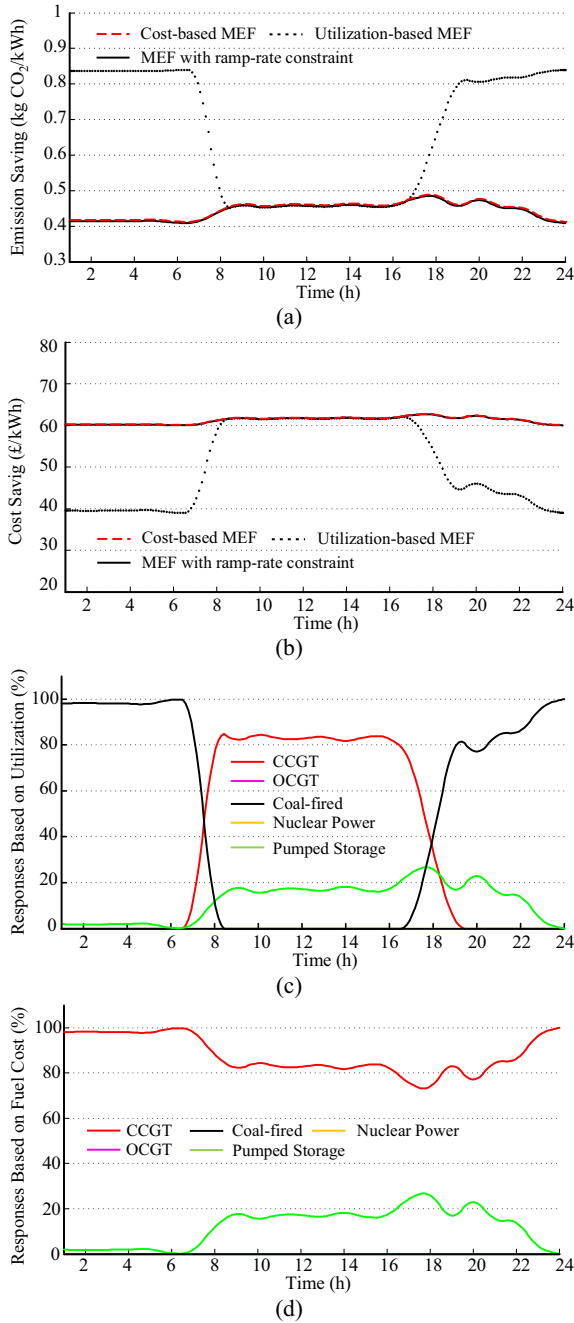


Fig. 14. Typical winter demand, 5% reduction, current fuel prices.

VII. CASE STUDY II: TYPICAL SUMMER SCENARIO

In a typical summer scenario, the maximum change for the coal-fired technology is 5256.98 MW/h, ramping up from 6852.99 MW to 9481.48 MW in half an hour. So the upper constraint of ramp rate for the coal-fired in a typical winter scenario is chosen as 5256 MW/h. Furthermore, two additional ramp rate scenarios (−5% and +5%) are also considered to see how MEFs can be changed with different ramping abilities. Fossil fuel prices are also based on the data from [13] and the sensitivity analysis of MEFs to current and future fuel prices is done in the section that follows.

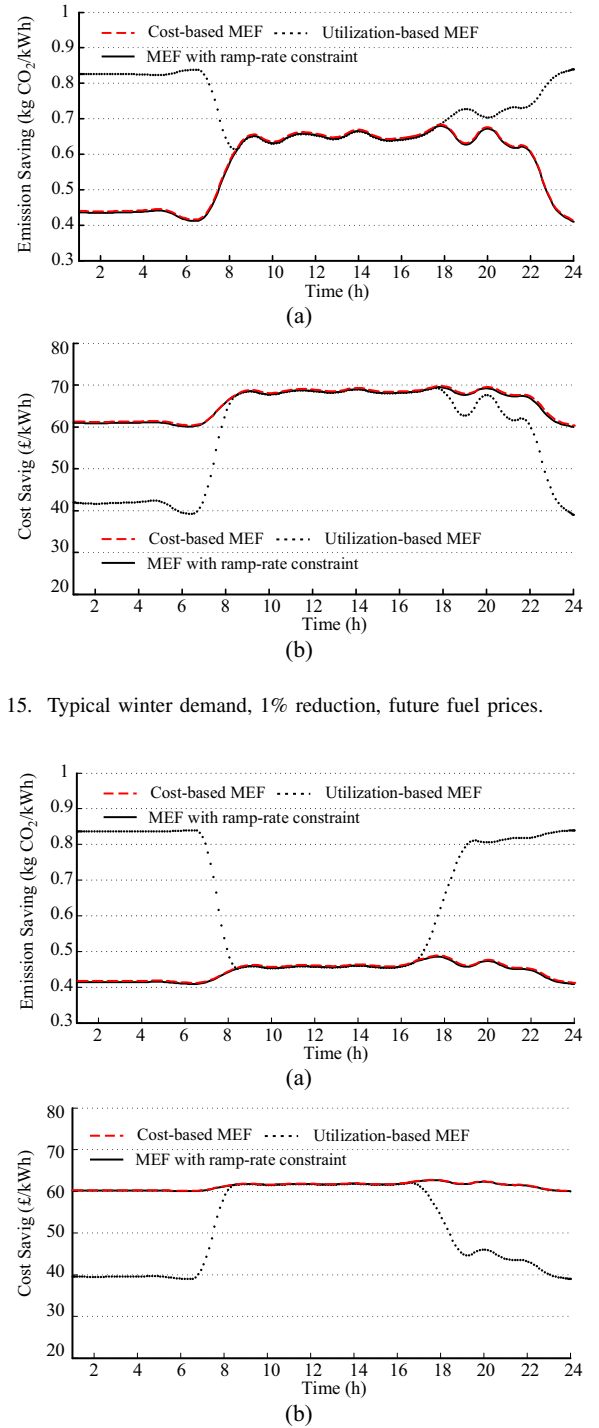


Fig. 15. Typical winter demand, 1% reduction, future fuel prices.

Fig. 16. Typical winter demand, 5% reduction, future fuel prices.

A. Impacts of Ramp Rate

First, MEFs without technical constraints are estimated in Fig. 17. In a typical summer scenario, when all hours of the day are expected to experience reductions at 1% level, most of the demand changes during peak hours are taken by the pumped units, as shown in Fig. 17 and Fig. 18. The utilization based MEF and the cost based MEF are the same throughout the day, holding MEF at around 0.85 kg CO₂/kWh during off-peak hours and 0.70 kg CO₂/kWh during peak hours. The

AEFs are about 0.5 kg CO₂/kWh, which are much lower than MEFs. In terms of cost savings, the MEF rate (around 10.5 £/MWh) is much higher than the AEF rate (around 8 £/MWh), because the marginal plants are generally more expensive than the base-load plants.

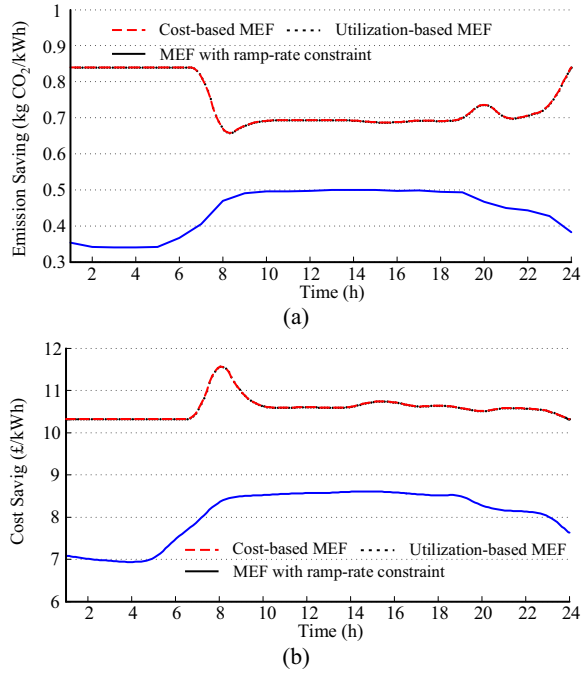


Fig. 17. Typical summer demand, 1% demand reduction.

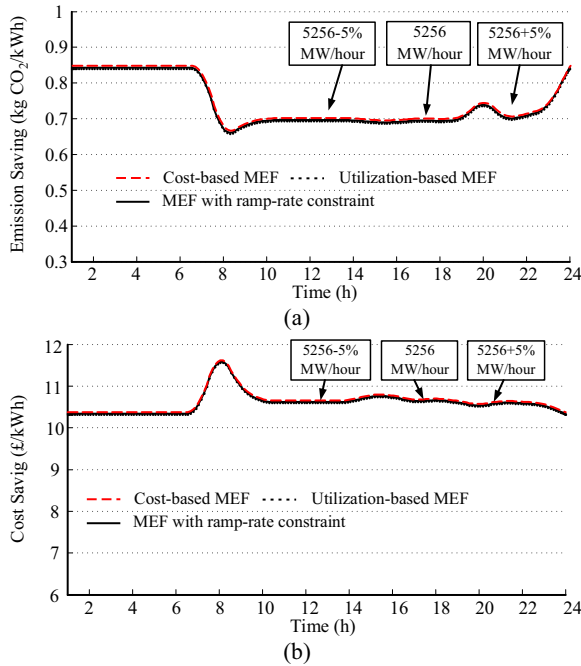


Fig. 18. Typical summer demand, 1% reduction with ramp-rate constraint.

When the demand reduction is increased up to 10% (Fig. 19), more coal-fired plants are needed to respond to demand changes. However, during off-peak hours, the coal-fired plants online are insufficient to meet the 10% reduction, thus trig-

gering some CCGT units to respond. As shown in Fig. 20(c) and Fig. 20(d), around 25% of demand reductions are taken by the CCGT units during the off-peak hours.

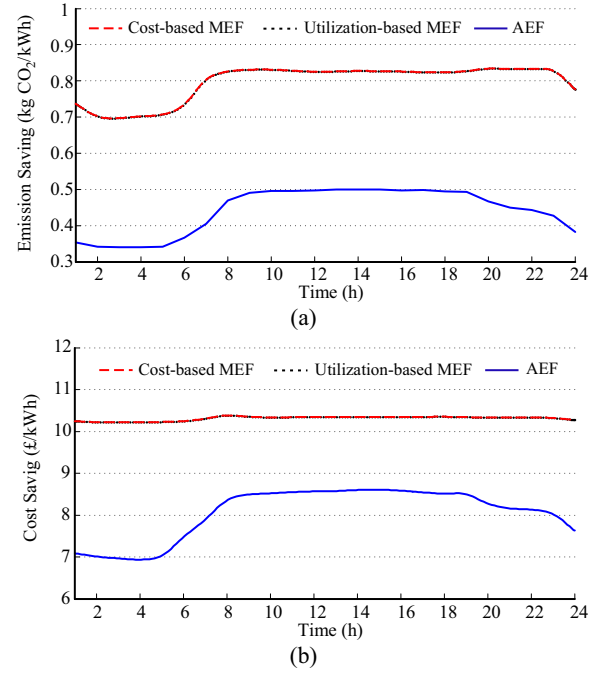


Fig. 19. Typical summer demand, 10% demand reduction.

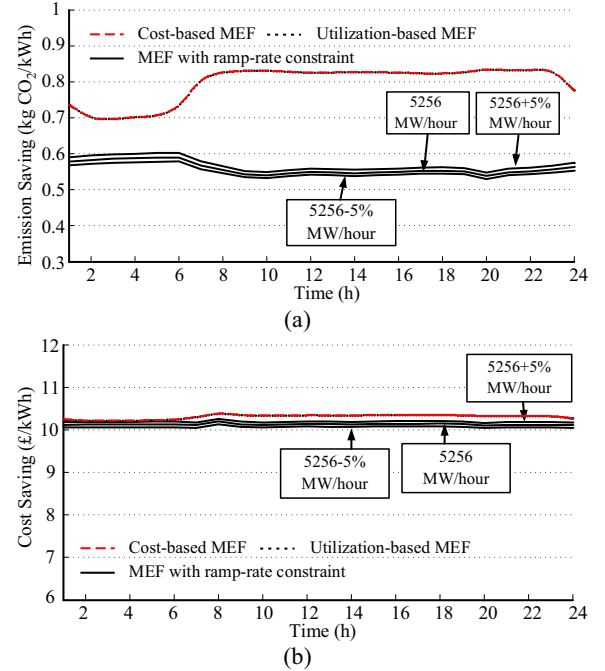


Fig. 20. Typical summer demand, 10% reduction with ramp-rate constraint.

However, if considerations of ramp-rate constraints are included in the MEF estimation (shown in Fig. 20), the coal-fired generators are constrained. Part of the demand changes have to be transferred to CCGT units, depending on how much the coal-fired power is constrained. So the MEFs turn out to be a coalition of the two technologies.

From these three levels of demand reductions (1%, 5%, and 10%), it can be seen that both the utilization based MEF and the cost based MEF are higher than the system average emission rate in all scenarios. If constraints of ramp rate are considered in the estimation, the MEFs are lowered when demand changes are large enough to affect the assumed merit order. Results confirm the conclusion that when demand reduction is significant (10% in this scenario), the ramp rate can have an important impact on MFE estimation and consideration of ramp-rate constraint becomes indispensable.

B. Different Fuel Prices

In order to see how the MEF estimation might be affected with different fuel prices in the summer scenario, the current prices scenario and the future prices scenario are also used in this study. Results of relevant MEFs are shown as follows.

When fuel prices are replaced with current prices (shown in Fig. 21), the merit order reshuffles and CCGT units replacing the coal-fired become the marginal plants. Fuel cost based MEFs decrease as CCGT is a cleaner technology than coal. Such a replacement is even clearer when the demand reduction is set to be 10% (shown in Fig. 22).

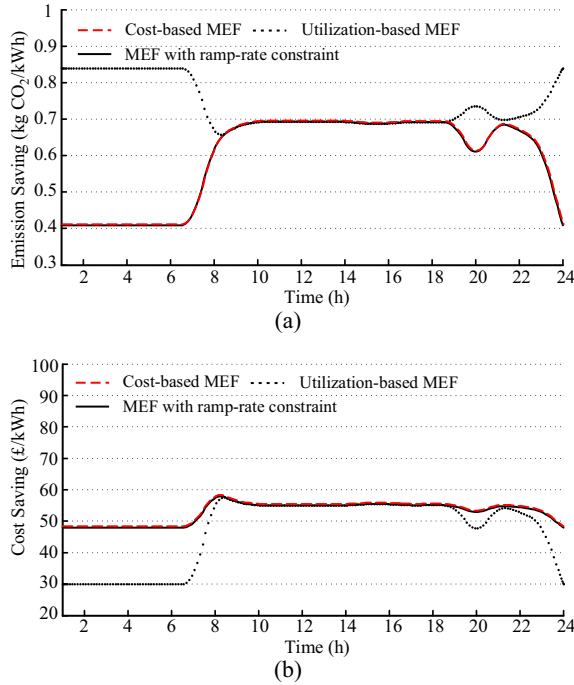


Fig. 21. Typical summer demand, 1% reduction, current fuel prices.

When future fuel prices are used to set up the merit order (shown in Fig. 23 and Fig. 24), estimated MEFs are similar to current price MEFs as the responding generator is the same as the current scenario.

As mentioned earlier, the fuel cost based MEF is determined by the fuel prices. Based on the projections of the DECC, the coal price will increase by 29.47% until 2032 and the gas price will increase by 23.33%. However, coal will be cheaper than gas in terms of fuel price. That is the reason why the current scenario yields similar results as the future scenario. This is

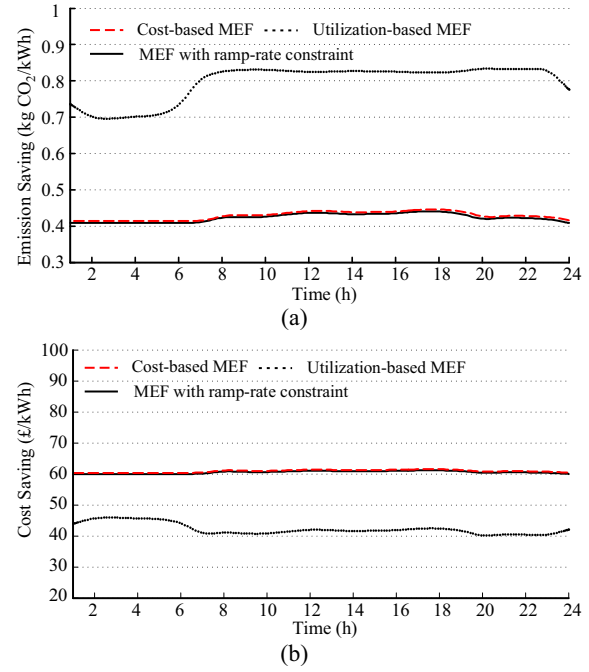


Fig. 22. Typical summer demand, 10% reduction, current fuel prices.

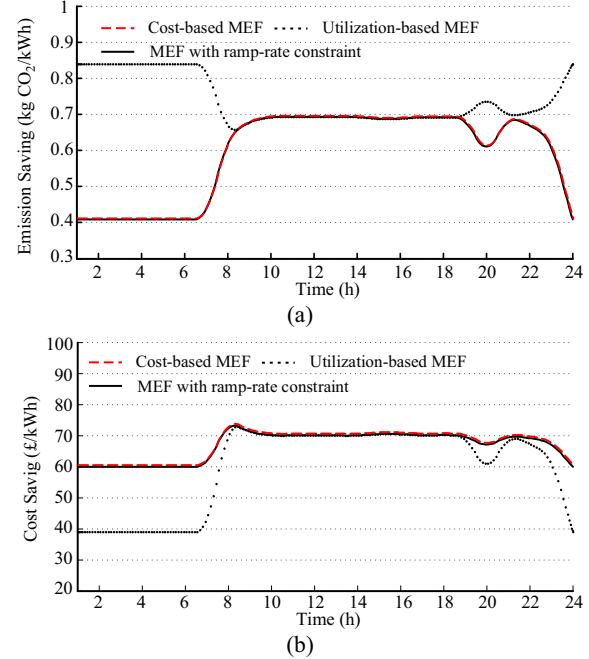


Fig. 23. Typical summer demand, 1% reduction, future fuel prices.

also the reason why the marginal emissions savings are not sensitive to price changes from this point in time, even under a fuel cost based merit dispatch.

The analyses of summer scenarios confirm the conclusions made by winter scenarios, namely, 1) fuel cost based MEFs are significantly changed between the past scenario and the current scenario, 2) the ramp rate can have a big impact on the MFEs' estimation especially when the demand changes are significant, and 3) consideration of ramp-rate constraint is indispensable in MEF estimations.

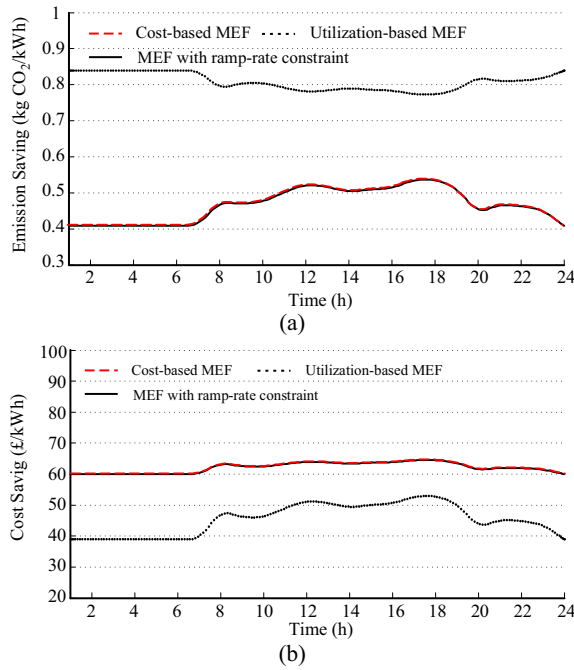


Fig. 24. Typical summer demand, 5% reduction, and future fuel prices.

VIII. CONCLUSION

This paper proposes an improved MEF estimation model by considering ramp-rate constraint in fuel cost based merit order dispatch. It is applied to a British power system to illustrate the differences between AEF and MEF in GB, and comparisons with two conventional merit order approaches of estimating MEF (utilization level based and fuel cost based) are also made. A sensitivity analysis of MEF to a few price scenarios is conducted to see how MEF estimation can change with different fuel prices. Conclusions are drawn as follows:

- 1) MEF demonstrations in the GB power system show that MEF can be lower than AEF when low carbon generators work on the margin and are triggered by DSR. Application of AEF to marginal impact assessment can lead to significant underestimation of potential emission savings.
- 2) Case studies show that MEF displays a high degree of variability over the course of a year as well as over the course of a day. Therefore, a fixed marginal factor for both peak and off-peak times is not justified, i.e., a single marginal emission cannot provide adequate reflection of the potential carbon savings from DSR for each time period during a day.
- 3) When demand reduction is small, the impact of ramp rate on generators on the estimation of MEF is trivial as the flexible generator has enough capability to balance the reduction. However, if the demand changes are larger than the ramping capability of flexible generators, such as 5% and 10% in the demonstration, the ramp-rate constraint has a major impact on MEF results, and considering ramp-rate constraint in the MEF estimation becomes indispensable.

- 4) The fuel-cost-based MEF is subject to the fuel prices being considered. The MEF in past price scenarios has higher carbon emissions savings than the current scenario. However, for the future price scenario, fuel cost based MEF turns out to be the same as the current price scenario, as gas prices are always expected to be more expensive than coal.
- 5) According to the case studies, there is a potential for increasing MEF rates by triggering heavy carbon polluters working on the margin, which illustrates the importance of introducing carbon mechanisms to the power sector. If the carbon prices introduced can make low carbon technology more competitive in generation, then the environmental benefit using DSR can also be enhanced.

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